1 Score the current domain

No trial cavity



$$\widehat{D}_T(z) = -\operatorname{avg}_{i \neq j}(U_i - h_z)P_j$$

Rank feasible, separated candidates

2 Freeze a finite proposal

Fix center and radius

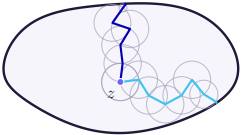


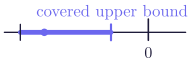
Choose the proposal before validation

The dashed circle is not yet a hole

3 Check with fresh samples

Account for all error terms



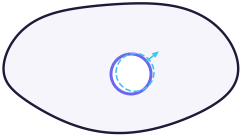


Pass only when the upper bound is < 0

Fail: resample under a valid schedule, or skip

4 Insert, then refine

Only after the test passes



$$\Omega \leftarrow \Omega \setminus K_{\varepsilon,z}$$

Optional Differential WoS shape step

General-objective guard

$$\widehat{D}_T + r_N(\delta) + b_{\text{term}} + b_{\text{cache}} + C_z \rho_z(\varepsilon)^\alpha < 0$$